FLOWMLLAB | INCREMENTAL LECTURE 7.1

# Rarefied hypersonic cylinder

## Operator learning, baselines, and honest UQ

From author-supplied DSMC fields to a leakage-controlled teaching experiment

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Companion: notebooks/week07_1/W7_1_Hypersonic_Rarefied_Cylinder_DeepONet.ipynb

# What this increment adds

OUTCOMES

- Separate freestream Mach from the predicted local Mach field.
- Formulate a parameter-to-field operator for DSMC cylinder data.
- Freeze whole-case splits before model selection.
- Read the branch, trunk, layerwise fusion, and output head.
- Challenge the neural model with a structured interpolation baseline.
- Audit training error; a single MLP does not provide calibrated uncertainty.

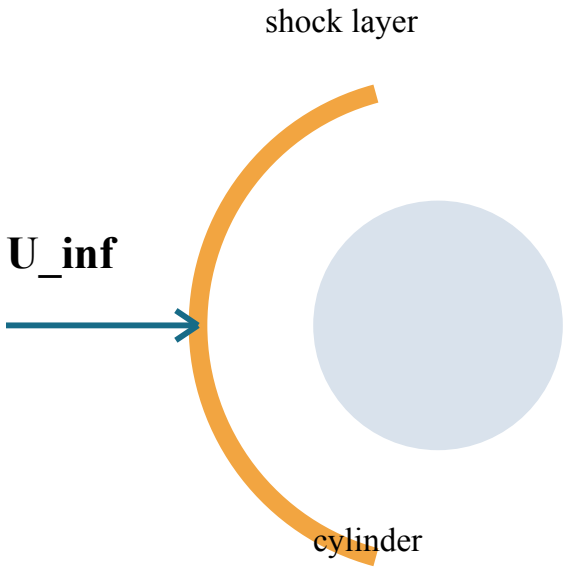
CLAIM

The lab executes a compact, author-released DSMC derivative. The default CPU model is a **teaching analog**; it does not reproduce the paper's full-resolution accuracy or speedup.

RULE

**If a transparent baseline wins, retain that result. Complexity is not evidence.**

# Physics: three scales, three questions




**Mach number:** M_inf = U_inf/a_inf controls compressibility. The target M(x,y) is a different, spatial field.

**Knudsen number:** Kn = lambda/D measures rarefaction. As lambda becomes non-negligible, continuum closure and no-slip assumptions need qualification.

**Sampling scale:** DSMC fields estimate particle moments. A sharp shock and finite particle count make target noise spatially nonuniform.

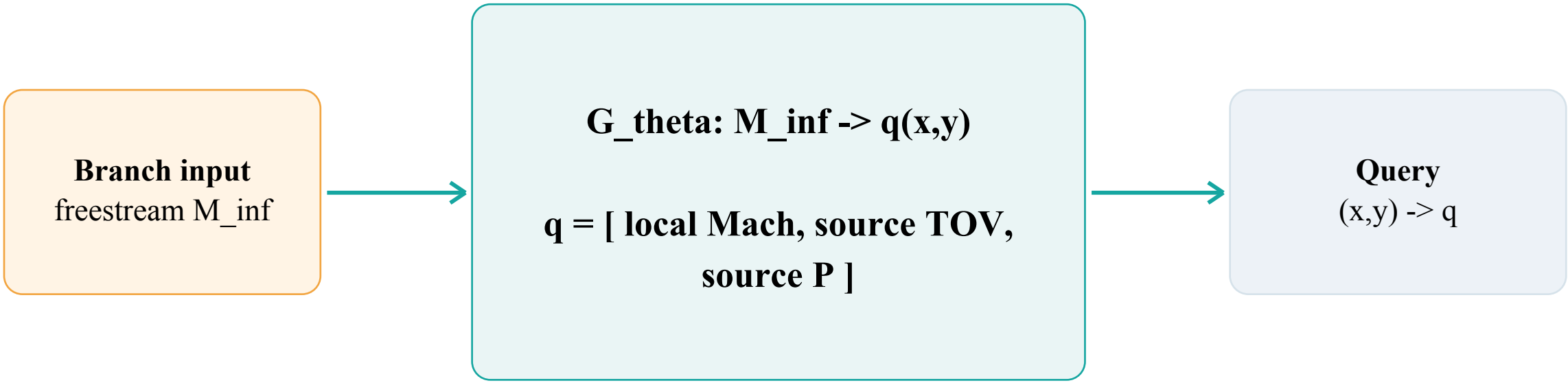
# Mach independence and the classroom MLP

**Physical motivation.** At high Mach, appropriately nondimensionalized fields can vary slowly with Mach under compatible gas, rarefaction, geometry and boundary conditions. This motivates interpolation; a small error alone is not proof of Mach independence. The source TOV/P scaling must be verified.

**Required model.** Three 96-unit tanh layers; inputs (Mach, x, y); equal standardized-target MSE. Scalers use training cases only. Adam, fixed seed 760, maximum 300 epochs and 40-epoch patience; best epoch selected on four whole validation cases. Report training error as an underfitting diagnostic.

**Provenance.** Research DSMC data precede the course. This MLP is a new teaching model, not the paper's trained Fusion-DeepONet. A single fit provides no calibrated UQ. The committed dataset is CC BY 4.0, authorized by Ehsan Roohi on 2026-09-05; see DATA_LICENSE.md.

# The learning object is an operator




**A field surrogate must be judged at unseen flow cases, not merely unseen grid points from a case already used during training.**

# Data audit: small derivative, complete provenance

| 20 | 400 x 400 | 50 x 50 | 44,500 | 0.5 MB |
|---|---|---|---|---|
| Mach cases | source grid per case | deterministic selection | finite retained points | committed NPZ |

**KEPT**

- M_inf, x, y
- local Mach, source TOV, source P (legacy ratio keys do not verify scaling)
- case ID and original source-grid row
- archive and derivative SHA-256 hashes

**EXCLUDED**

- 1.4 GB source ZIP and 4.9 GB expanded files
- training logs and duplicate checkpoints
- historical scripts with inconsistent protocols
- research checkpoints not needed for the lab

# Whole-case splitting prevents a false claim

| TRAIN | VALIDATION | BLIND INTERP. | BLIND EXTRA. |
|---|---|---|---|
| 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 | 8.25, 8.75, 9.25, 9.75 | 5.5, 6.5, 7.5, 8.5, 9.5 | 15 |

**Wrong question:** can the network fill random points from a Mach case it has already seen?
**Scientific question:** can it predict a complete field at an operating condition excluded from fitting and tuning?

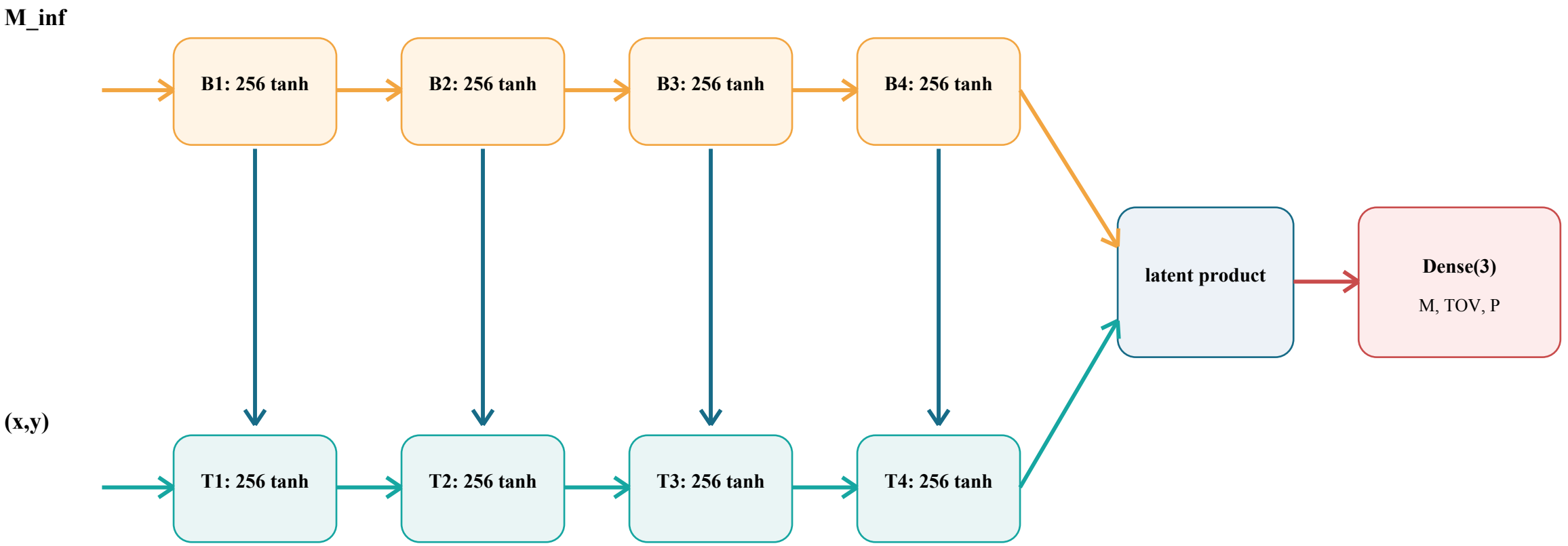
# The mandatory baseline is field interpolation

$$q_hat(M_q) = q(M_a) + [(M_q - M_a)/(M_b - M_a)]\ [q(M_b) - q(M_a)]$$

| Split | Local Mach | Temperature | Pressure |
|---|---|---|---|
| **Blind interpolation** | **0.398%** | **0.609%** | **0.779%** |
| Mach 15 extrapolation | 0.538% | 0.877% | 0.848% |

**Result: a neural operator must beat a sub-1% structured baseline here, or justify itself through a different deployment constraint.**

# Educational multi-output Fusion-DeepONet

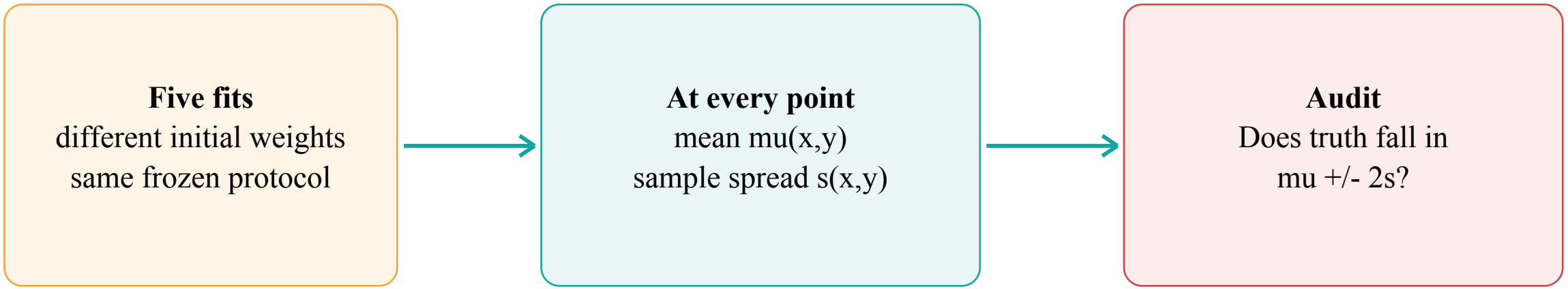



The corrected teaching head keeps the latent product vector before Dense(3). The legacy scalar-dot head forces three outputs onto one affine line; legacy_scalar is compatibility-only. This builder does not verify paper-checkpoint equivalence.

# Training protocol: every choice changes the claim

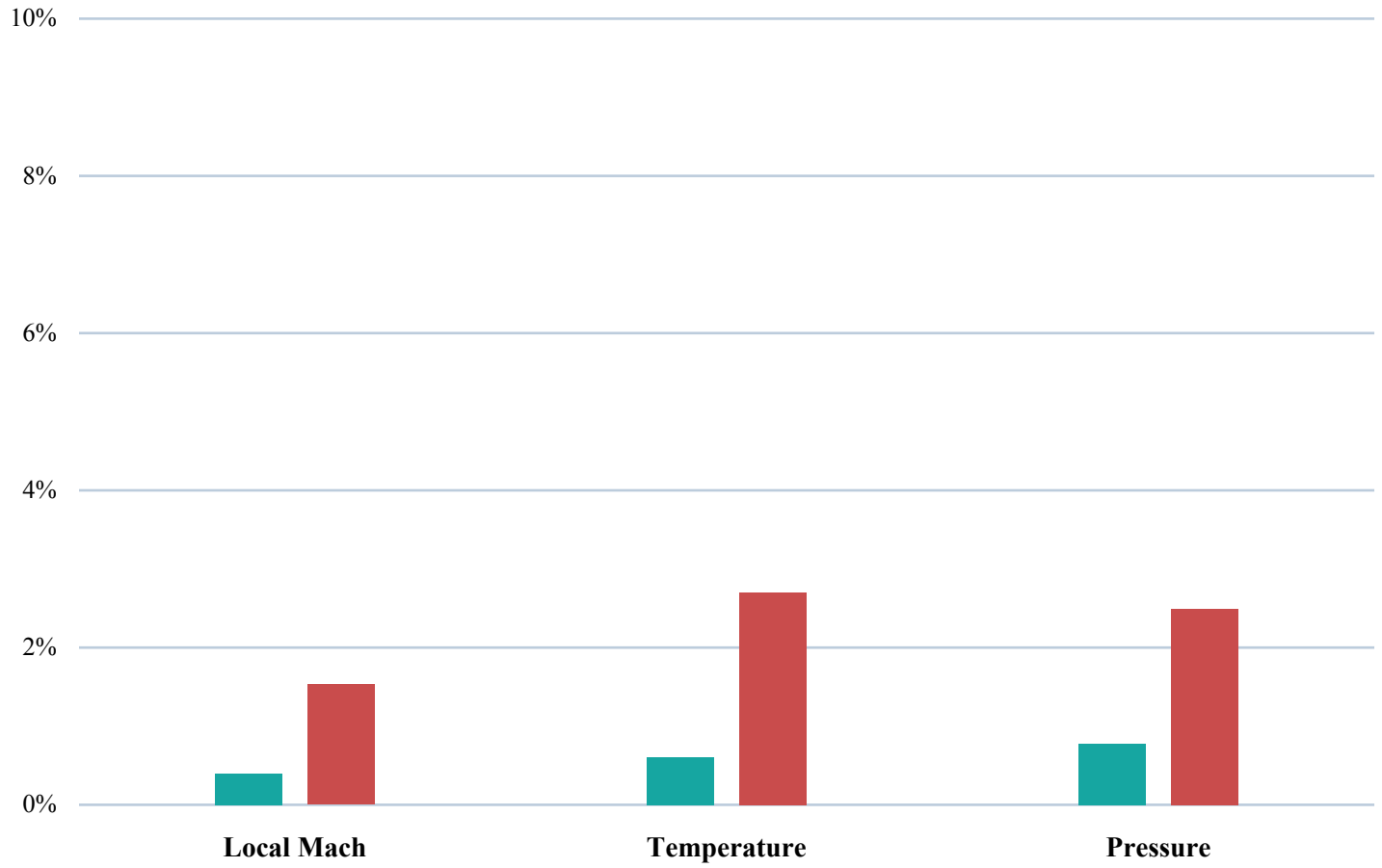
| | | |
|---|---|---|
| 1 | **Fit scalers** | Training cases only; MinMax inputs and standardized targets. |
| 2 | **Optimize** | Adam with validation-monitored early stopping and learning-rate reduction. |
| 3 | **Weight** | Reviewed archive loss: MSE_M + MSE_T + 5 MSE_p in standardized space. |
| 4 | **Repeat** | Five independent initializations; preserve member predictions. |
| 5 | **Open blind cases** | After design is frozen; report interpolation and extrapolation separately. |

# Optional ensembles: not run by the single-MLP lab




**Low coverage means overconfidence. High coverage can also be uninformative if intervals are too wide. Report coverage and interval width, by target and by split.**

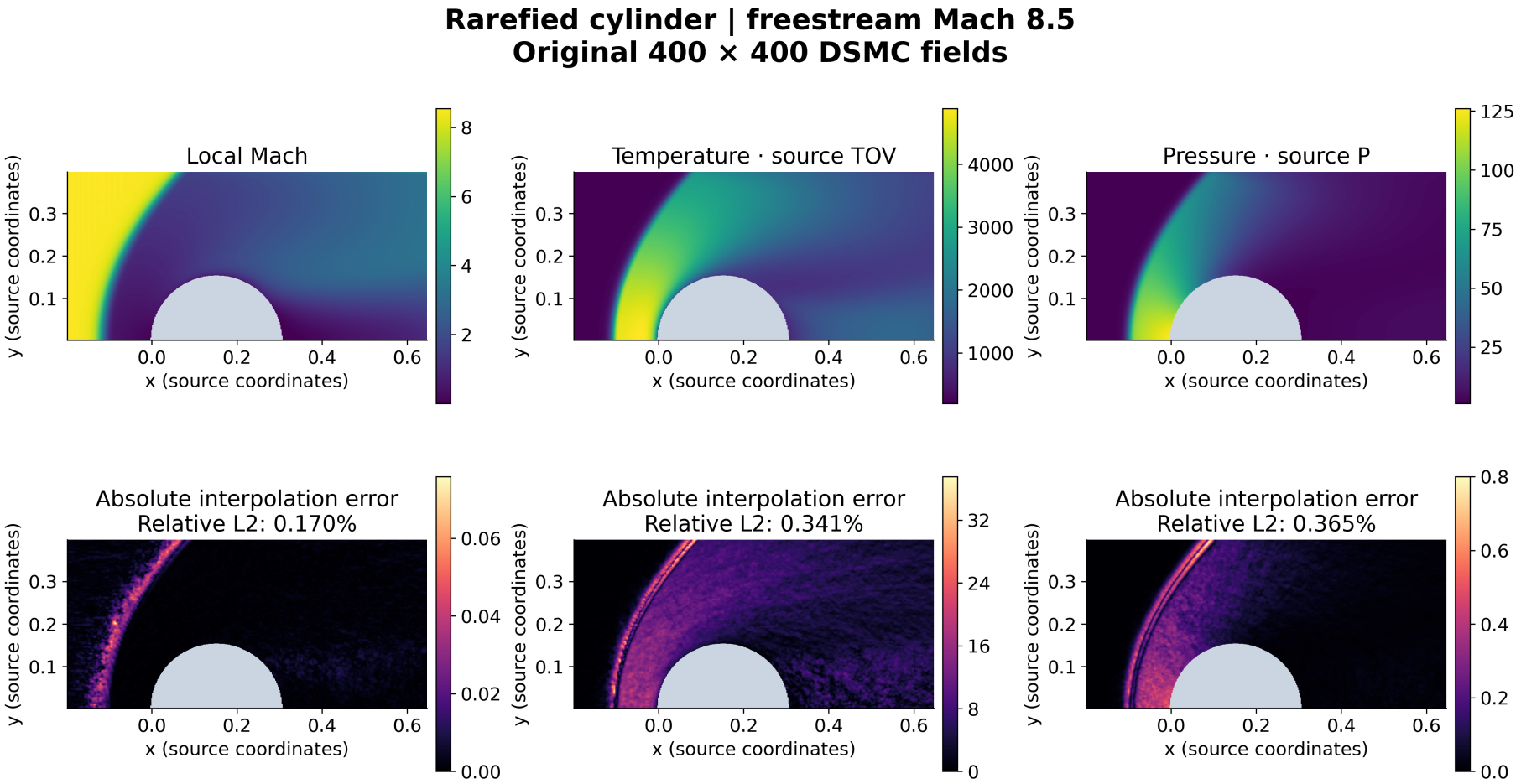
# Executed evidence: trained MLP versus interpolation




**Baseline**
0.398% | 0.609% | 0.779%

**Trained 3x96 MLP**
1.53% | 2.70% | 2.50%

# Full source-grid interpolation: separate from MLP metrics




Gray: masked solid/sentinel region. Interpolation uses Mach 8 and 9; no spatial smoothing.

# Decision gate: when is a neural operator justified?

### Accuracy
Does it beat field interpolation on frozen blind cases?

### Geometry
Must one model span grids, shapes, or boundary conditions?

### Availability
Will deployment lack bracketing high-fidelity solutions?

### Cost
Do many field queries amortize training and data production?

### Physics
Are shock location, surface loads, positivity, and limits acceptable?

### Uncertainty
Is spread calibrated on the deployment regime?

# Lab deliverables and claim boundary

**SUBMIT**

- Baseline and teaching-operator relative-L2 errors by target and split.
- One physics-based interpretation of an error hot spot.
- Training error and an explicit statement that the single MLP has no calibrated UQ.
- A short explanation of random-point leakage.
- One deployment condition that could justify full neural training.

**MAY CLAIM**

The compact DSMC derivative, frozen split, baseline, and CPU teaching analog were executed.

**MAY NOT CLAIM**

Published full-model accuracy, speedup, complete-field reproduction, or calibrated UQ.

# References and reproducibility trail


- Roohi et al. (2026), Neural Networks for Rarefied Gas Dynamics: Relaxation Problem, Polyatomic Shock Waves, and Hypersonic Cylinder Flow. Physics of Fluids 38, 057108. https://doi.org/10.1063/5.0334590
- Lu et al. (2021), Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. Nature Machine Intelligence 3, 218-229. https://doi.org/10.1038/s42256-021-00302-5
- Lakshminarayanan, Pritzel, and Blundell (2017), Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS 30.
- Bird (1994), Molecular Gas Dynamics and the Direct Simulation of Gas Flows. Oxford University Press.


Rebuild data: qa/build_hypersonic_cylinder_subset.py | Rebuild evidence: qa/run_hypersonic_cylinder_evidence.py | Source archive SHA-256 begins bda221759a37